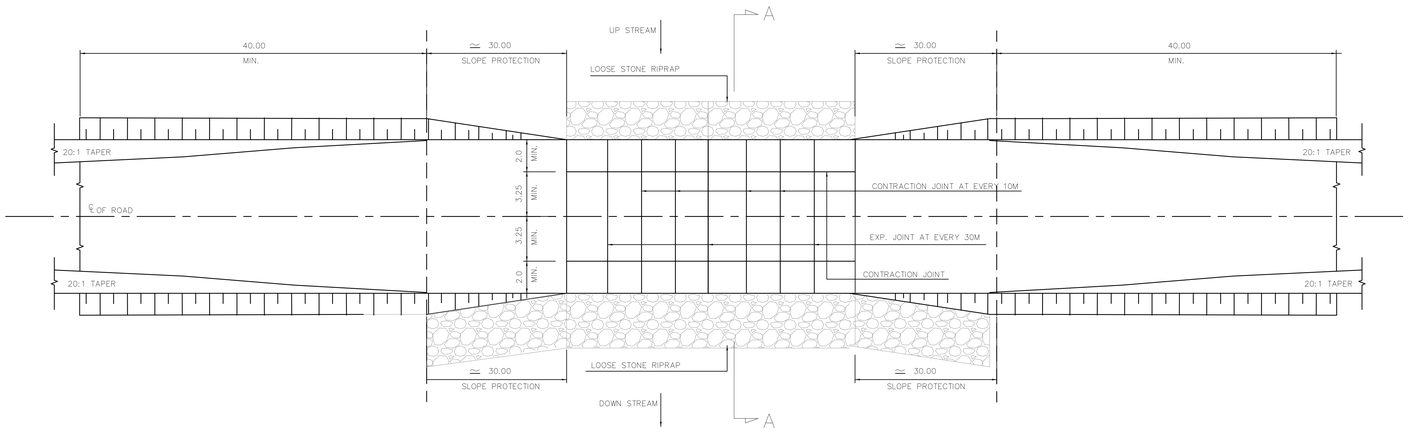
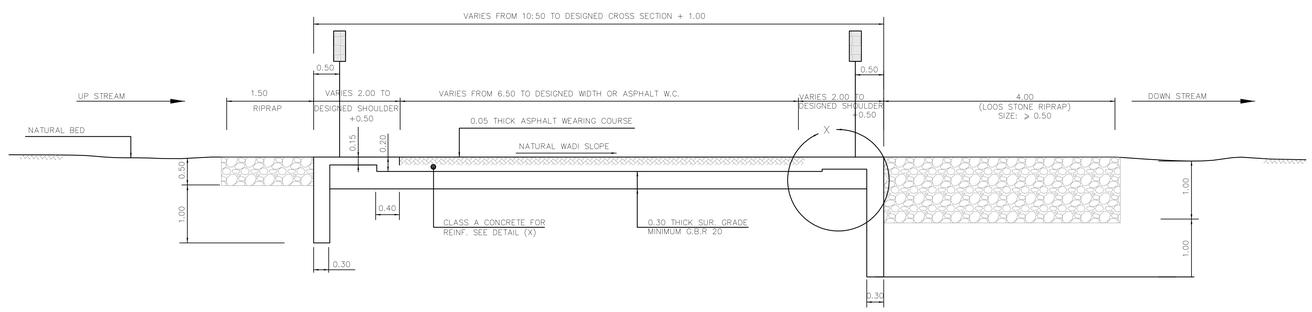
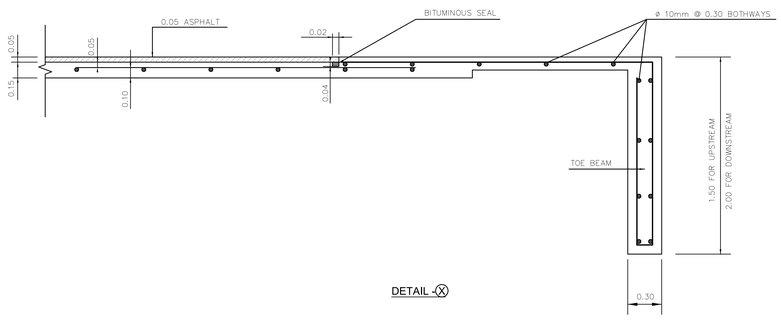


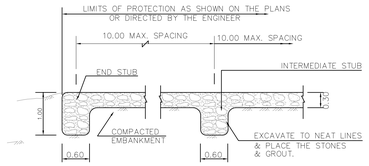
PLAN FOR IRISH CROSSING
TYPE 'A'



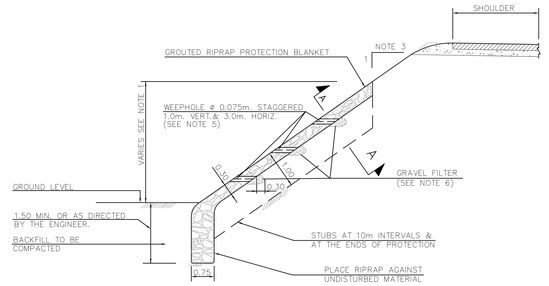
SECTION A-A



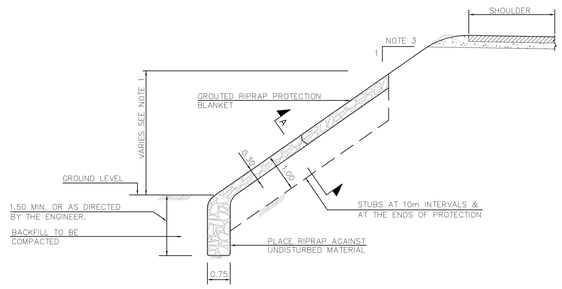
DETAIL -Ⓧ



SECTION OF STUBS



SECTION
GROUTED RIPRAP SLOPE PROTECTION WITH WEEPHOLES
FOR HEIGHT OF PROTECTION > 2.00m



SECTION
GROUTED RIPRAP SLOPE PROTECTION
FOR HEIGHT OF PROTECTION ⩾ 1.00m AND ⩽ 2.00m.

GENERAL NOTES:

1- ALL DIMENSIONS SHOWN ARE IN METERS.

2- TYPE & HEIGHT OF EMBANKMENT PROTECTION SHALL BE AS SPECIFIED ON PLANS.

3- FOR EMBANKMENT SLOPES SEE "STANDARD CROSS-SECTION DRAWINGS".

4- PROVIDE IN CONCRETE EMBANKMENT PROTECTION A CONSTRUCTION JOINT EVERY 6.00m & EXPANSION JOINT EVERY 12.00m. DETAILS OF THESE JOINTS WILL BE APPROVED BY THE ENGINEER.

5- WEEPHOLES ARE PLACED @ 0.5m. FROM TOE AND STAGGERED AT 1.0m. VERTICALLY AND 3.0m. HORIZONTALLY.

6- GRADATION OF THE FILTER MATERIAL SHOULD SATISFY THE FOLLOWING CRITERIA:
a) IN RELATION TO BASE MATERIAL RETAINED BEHIND THE FILTER
- D15f / D85b < 5
- 5 < D50f / D50b < 25
- D20f / D20b > 5
b) IN RELATION TO WEEPHOLES
- D85f / Dhole > 1-2

WHERE D15f, D20f, D50f AND D85f ARE USED TO DESIGNATE THE SIZE OF THE SIEVE THAT ALLOWS 15%, 20%, 50% AND 85% RESPECTIVELY BY WEIGHT OF THE FILTER MATERIAL TO PASS THROUGH IT. ALSO, D20b, D50b AND D85b ARE USED TO DESIGNATE THE SIZE OF THE SIEVE THAT ALLOWS 20%, 50% AND 85% RESPECTIVELY BY WEIGHT OF THE BASE MATERIAL TO PASS THROUGH IT. Dhole IS USED TO DESIGNATE THE DIAMETER OF THE WEEPHOLE IF ONE LAYER OF THE FILTER DOES NOT SATISFY THE A/M CRITERIA,

7-ALL CULVERT'S INLETS AND OUTLETS SHOULD
HAVE GROUTED RIPRAP OF SIZE d50=300 mm
EXCEPT THE OULET OF CW09 d50=400 mm.

8-MOT SPECIFICATION SHALL BE FOLLOWED
FOR THIS PROJECT

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		D01	04/02/2024	AS BUILT	SYA	SOR	AVG	SJM	
		C01	03/02/2024	ISSUED FOR CONSTRUCTION	SYA	SOR	AVG	SJM	
		A04	18/01/2022	REVISED AS PER EMPLOYER COMMENTS ON 11/08/2021	--	SNP	--	SJM	
		A03	19/07/2021	REVISED AS PER EMPLOYER COMMENTS ON 03/07/2020	--	MLI	--	SJM	
		A02	06/06/2020	REVISED AS PER EMPLOYER COMMENTS ON 13/05/2020	--	MLI	--	SJM	
		A01	16/04/2020	ISSUED FOR REFERENCE	--	SNM	--	SJM	

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DRAWING NUMBER	SA1930-PK1-DD-WS-502	REVISION	0
DATE	10/04/2020	DATE	11/04/2020	DATE	12/04/2020
DRAWN	Y.A	CHECKED	M.S	APPROVED	M.E

LARSEN & TOUBRO DRAWING NUMBER	019045_910_C_V_24019_004	REVISION 5
OUTOTEC DRAWING NUMBER	900707__GAU01__AG90__CTB17__00004	DOCUMENT ID OU100681672 / REVISION 5

DRAWN	--	--
CHECKED	SNM	13/04/2020
ENGINEER	--	--
APPROVED	SJM	16/04/2020
EMPLOYER		
SCALE	N.T.S	A1

MANSOURAH - MASSARAH GOLD PROJECT
DETAIL DRAWING
STORMWATER TYPICAL DETAILS-ACCESS ROAD TO MANSOURAH-MASSARAH PROCESS PLANT
SHEET 2 OF 3

DRAWING NUMBER MD - 216 - 49EA - EG - CV - DED - 1004
REV D01

MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

BECHTEL

Mansourah Massarah Gold Project
Comment Resolution Log Sheet / Compliance Report

Outotec
LARSEN & TOUBRO

PROJECT: MANSOURAH MASSARAH GOLD PROJECT **PROJECT NO.: 216**

STAGE: 4 - PROJECT EXECUTION. **REVIEW LOG DOC. NO. & REV.:**

ACONEX REF. NO. & DATE: BECHTEL-WTRAN-014220, 09 February 2022

STATUS CODE: A - APPROVED, C - APPROVED WITH COMMENTS, R - REVISE & RESUBMIT & O - OTHERS (INFORMATION, REFERENCE, REVIEWED)

DOCUMENT NUMBER	Area / Title	Comment No.	PMC/PMT COMMENT DATE: 09 February 2022	Comment by	EPC Response DATE: 03 February 2024	Status	Remarks
			General Comments (cc)				
MD-216-49EA-EG-CV-DED-1004-A04	49EA	1	- No hydraulic analyses have been included for supporting the design of Irish crossings. Hydraulic analyses are still required, using an appropriate software such as HEC-RAS, Flow-2D, or similar. For instance, it is said that the flow velocities are minimal at the crossing points, but this must be demonstrated with the results given by an appropriate hydraulic model. Standard designs offered by the MOT are limited to certain hydraulic parameters, therefore the EPC must confirm (through hydraulic modeling) that the specific hydraulic conditions of each crossing are adequate.	Bechtel	Incorporated in the revised drawing.	Closed	
MD-216-49EA-EG-CV-DED-1004-A04	49EA	2	Section 5 of the report cited provides the rationale for computing the design flows using the software HEC-HMS, which is fine. Section 6 “Hydraulic Design” does not provide sufficient background and supporting information for reaching the design conclusions given in Table 12 for Irish crossings. There is one appendix showing hydraulic analyses for culvert crossings only. Hydraulic analyses for Irish crossings are still required.	Bechtel	Incorporated in the revised drawing.	Closed	